CS3: Case Study Create - Deepfake Image Detection

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Case study repository: https://github.com/angadv12/KALz_Project_3

Hook Document

Can You Catch an AI Image Before It Spreads?

Imagine the Cavalier Daily receives two images tied to a breaking campus story. One came from a real photo source. The other came from a generative model. Both look believable at a glance, and the newsroom needs an evidence-based call before the image moves across group chats and social media.

You are the analyst on that first pass. Your job is not to build a perfect truth machine. Your job is to test whether a reproducible computer-vision workflow can separate real images from AI-generated images better than a person scanning pixels by eye.

The KALz project gives you a starting point: the CIFAKE dataset, a clean repository, two pretrained CNN models, saved evaluation outputs, and Grad-CAM examples. CIFAKE contains 120,000 low-resolution images split evenly between REAL images from CIFAR-10 and FAKE images generated to match that style. The original project compared ResNet-50 and DenseNet-121. Both models beat the 80 percent target and the 92.98 percent CIFAKE paper baseline, with DenseNet-121 reaching 96.97 percent test accuracy.

Your mission is to turn that project into a short, honest technical brief for a non-expert reader. You will explain the problem, run or inspect the workflow, compare the models, and decide what the results do and do not prove. The strongest answer will pair metrics with skepticism: confusion matrices show where the model failed, and Grad-CAM helps you ask whether the model focused on meaningful image regions or dataset-specific artifacts.

By the end, you should be able to answer one practical question: if someone handed you a suspicious image, what evidence would make you trust or doubt a CNN-based detector?

Rubric for the Student Deliverable

Target student: second-year UVA student. Deliverable: a 3-4 page technical brief, saved as a PDF, plus a GitHub repository that lets another student inspect the code, data instructions, model outputs, and references.

Spec Category	Spec Details
Formatting and submission	• You submit a GitHub repository link and a final PDF brief. • Your repo contains the brief, notebook or scripts, output figures, metric files, and reference list. • Your brief uses clear headings, labeled figures, and citations. Do not bury results in raw notebook output.
Problem framing	• You explain why AI-generated images create a trust problem in plain language. • You state the research question: which pretrained CNN, ResNet-50 or DenseNet-121, detects CIFAKE images more effectively? • You define REAL and FAKE without assuming the reader already knows CIFAKE.
Data understanding	• You describe CIFAKE's size, class balance, train/test split, and image source. • You include at least one EDA point from the repository, such as class balance, image size, brightness, or pixel variance. • You name one dataset limitation, especially the 32x32 resolution and the single fake-image generation setup.
Preprocessing and model workflow	• You describe the path from raw images to model-ready tensors: resize to 224x224, normalize with ImageNet values, and batch the data. • You explain why transfer learning helps here. • You compare ResNet-50 and DenseNet-121 by their design idea, not by name alone: skip connections for ResNet, dense feature reuse for DenseNet.
Evaluation and comparison	• You report accuracy, precision, recall, and F1 for both models. • You compare the models to the 80 percent project target and the 92.98 percent Bird and Lotfi baseline. • You use the confusion matrices to discuss false positives and false negatives instead of saying only that the models were accurate.
Grad-CAM and interpretation	• You include at least one Grad-CAM example or a written interpretation of the provided Grad-CAM outputs. • You explain what Grad-CAM can suggest and what it cannot prove. • You connect interpretability back to the original question: would the detector help a human make a better judgment?

Spec Category	Spec Details
Conclusion and limits	• You make a clear claim about which model performed best on this dataset. • You explain why 96-97 percent accuracy on CIFAKE does not guarantee real-world deepfake detection. • You propose one next test, such as a higher-resolution dataset, a newer generator, or a vision transformer baseline.

Materials, References, and Submission Notes

Repository Materials

- README.md: setup path, repository map, Colab workflow, final metrics, and references.
- scripts/01_EDA.py: local EDA for class balance, samples, image size, brightness, and pixel variance.
- scripts/02_03_04_preprocess_training.ipynb: preprocessing, ResNet-50 training, DenseNet-121 training, and checkpoint saving.
- scripts/05_06_07_eval_gradcam_report.ipynb: test evaluation, confusion matrices, Grad-CAM, and summary reporting.
- output/evaluation/: classification reports, prediction files, confusion matrices, and metrics_summary.csv.
- output/gradcam/: model-specific Grad-CAM example figures.

Suggested Hard-Copy Sources

- Explainer: Google DeepMind, Identifying AI-generated images with SynthID. This gives students a readable entry point into AI image identification and watermarking.
- Technical article: Bird and Lotfi, CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images. This explains the dataset and baseline that the project uses.

References

- Bird, J. J., and Lotfi, A. (2024). CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images. IEEE Access. doi: 10.1109/ACCESS.2024.3356122.
- CIFAKE dataset. Kaggle: birdy654/cifake-real-and-ai-generated-synthetic-images.
- Google DeepMind. Identifying AI-generated images with SynthID.
- He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep Residual Learning for Image Recognition. CVPR.
- Huang, G., Liu, Z., van der Maaten, L., and Weinberger, K. Q. (2017). Densely Connected Convolutional Networks. CVPR.
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. (2017). Grad-CAM. ICCV.

Print Packet Checklist

- Place the one-page hook document on the left front of the manila folder.
- Place the student rubric behind the hook document.
- Print the SynthID explainer and the CIFAKE article or article landing page for the right side.
- Submit the GitHub repository link on Canvas. The physical folder still needs to go to SDS room 436.